6th International Conference on Advancement in Electronics & Communication Engineering 2026 : Submission (173) has been created.

1 message

Microsoft CMT <noreply@msr-cmt.org> Mon, Apr 27, 2026 at 16:36
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Hello,

The following submission has been created.

Track Name: Track 10:Emerging Trends in Large Language Models (LLMs) and High-Performance Computing (HPC) Systems

Paper ID: 173

Paper Title: Experimental Performance Evaluation of Monolithic and Microservices Architectures Using Docker-based Deployment

Abstract:
Software architecture plays a vital role to a significant extent in determining the scalability, performance, and maintainability of modern-day applications. Comparatively, monolithic vs. microservices architecture type architecture are both very widely used; however, there have not been any empirical studies published comparing the benchmarking of these two architectures (monolithic versus microservices) and their performance characteristics.

The purpose of this paper is to provide a comparative study of implementations-based monolithic versus microservices architectural style through experimental analysis by deploying an actual production-like implementation of each architectural style in a real-world application environment. To achieve the above, we created two functionally equivalent applications remotely from a monolithic application to a microservices-based application using docker or containerized deployment of each service (using the container orchestration) as the deployment mechanisms.

A synthetic workload data set was created to allow evaluation of the performance of both systems against each system's workload through metrics of user engagement, response time, throughput, cpu utilization, and deployment efficiency. Our experiments showed that, while the performance of the monolithic system was lower at lower workload levels (because there was less communication), microservices based system exhibited both greater scalability and fault isolation at high workload levels.

The major contribution of this research provides an experimental environment controlled of both architectural types of systems using quantifiable metrics for developers to make more informed decisions on which type of system will meet the requirements of their systems. The findings of this study illustrated the trade-offs between the simplicity of system architecture vs. scalability of system architecture types.

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Submission Questions Response:
1. Author Consent
   Agreement accepted
2. Institute & Place of Study
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3. Confidence of Paper
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Thanks,
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